Robust Image-Based Visual Servo Target Tracking of UAV with Depth Camera

Huazhang Chen
School of Automation
Beijing Institute of Technology
Beijing 100081, China
Email: 3220222237@bit.edu.cn

Kewei Xia
Advanced Research Institute of Multidisciplinary Sciences
Beijing Institute of Technology
Beijing 100081, China
Email: kwxia@bit.edu.cn

2024 IEEE International Conference on Industrial Technology (ICIT) | 979-8-3503-4026-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICIT58233.2024.10540793

***Abstract*—Tracking a target by using an unmanned aerial vehicle (UAV) without prior knowledge of the target's velocity and position is a critical and challenging task. This paper proposes a feasible approach that combines image-based visual servo (IBVS) with robust control. More specifically, a target detection algorithm is employed to acquire and calculate the pixel coordinates of the target's center. Simultaneously, the UAV utilizes a depth camera to measure the depth distance between the target and the UAV. Subsequently, the UAV directly computes flight control commands at the image pixel level through robust control, which comprises a force controller for position and a torque controller for attitude, separately. The UAV can continuously track the target through the aforementioned procedures. To validate the reliability of the proposed approach, a series of experiments and simulations are conducted.**

***Index Terms*—unmanned aerial vehicle (UAV) , image-based visual servo (IBVS), robust control, target tracking**

I. Introduction

The tracking and detection of target by using a UAV poses a crucial and formidable challenge [1][2], particularly within the environment where the global positioning system (GPS) is proved inapplicable [3]. Nevertheless, the image is susceptible to natural conditions, including the UAV's attitude and external environmental factors such as wind speed and lighting. To mitigate this impact, a practical approach is proposed, which pays attention to image-based visual servo (IBVS) based on robust control [4] as a feasible remedy.

As a critical control method for robotic systems, visual servo control employs visual information as feedback for non-contact environmental measurements. Visual servo control is classified into two types [5]: Position-Based Visual Servo (PBVS) and IBVS. Due to the inability of PBVS to adapt to complex environments, UAV usually employs IBVS for tasks. Guenard *et al.* [6] proposed an IBVS control after considering the advantages of IBVS and PBVS. Shi *et al.*[7] proposed an IBVS method with a reinforcement learning method, Q-Learning for adaptive servo gain adjustment. Miranda-Moya *et al.* [8] proposed A sliding mode control based on IBVS for target tracking under perturbations. PBVS requires controller design in a three-dimensional Cartesian space, which necessitates camera coordinate transformation calibration. Due to its primary reliance on pose information, PBVS typically involves lower computational loads, making it a common choice for UAV target tracking tasks. However, this strategy demands high model accuracy. Even slight errors in image measurements can lead to attitude inaccuracies. Moreover, PBVS is susceptible to occlusions, disturbances, and requires recalibration of camera intrinsic parameters in varying environments.

In contrast to PBVS, IBVS directly defines the control errors as the difference between current and desired image features. This approach eliminates the need for precise system modeling and simplifies the controller. IBVS is less sensitive to camera and sensor calibration errors, ensuring better maintenance of the target within the camera's field of view (FOV). Therefore, IBVS performs better in complex situations where continuous and rapid acquisition of target information may not be feasible.

Deep learning has made significant strides in recent years, especially within the domain of convolutional neural networks. Nowadays, deep learning algorithms have led to the development and packaging of models. Therefore, detection is not the central focus of our study. We have chosen the established YOLOv3 target detection algorithm [9].

In this work, we introduce an approach based on IBVS. Initially, camera information is transmitted to a ground station via network protocols, enabling the UAV to identify the target, calculate the relative position and velocity between itself and the target. Subsequently, the image recognition problem evolves into a real-time tracking issue. Then, a robust control strategy [4] is referenced and combined with IBVS. Finally, the proposed approach is validated through simulations and practical experiments in both simulated and real-world environments. The main contributions are summarized as follows:

1. Introducing an approach that integrates depth distance and IBVS, this approach utilizes a straightforward structure and relies on a single set of estimated parameters, excluding the need for target position and velocity information.
2. A comprehensive series of experiments, encompassing Gazebo simulations and practical experiments, highlight the feasibility of the controller.

The structure is presented as follows: Section II outlines the recent target tracking problem and the primary framework of UAV and camera. Section III presents the primary contributions of this approach, encompassing IBVS model, robust

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on March 30,2026 at 23:06:09 UTC from IEEE Xplore. Restrictions apply.

controller design. Section IV presents the validity of the strategy by simulations and practical experiments. Section V summarizes the conclusion.

II. Problem Description

In contrast to traditional PBVS tracking approaches, we propose an IBVS control law for the UAV and the depth camera by directly using image information, eliminating the need for the estimation of the camera's three-dimensional parameters in F_C. Since the camera is fixed above the UAV, the three Euler angles of the camera relative to the UAV are all $0°$. Consequently, due to the coupling of the UAV and the camera, the camera frame coordinate and UAV body frame coordinate coincide, indicating that the rotation matrix is an identity matrix.

However, there are two challenges in this work:

- How to measure depth distance.
- Minor perturbations in the UAV's attitude can lead to considerable variations in the target's position in the image coordinate. During mission execution, the UAV must precisely track the target's pixel center. Consequently, accomplishing tracking in this situation presents a challenge.

A. Relevant Dynamics Model

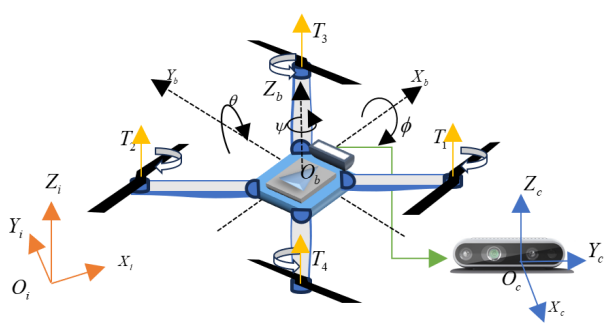


Fig. 1: Coordinate frames

The relative reference frames are constructed as depicted in Fig.1. The camera is positioned above the flight controller, and a constant transformation matrix, derived from their relative position and attitude, links the camera frame coordinate and UAV body frame coordinate. To simplify the complexity of the issue, the axes of the camera and UAV body frame coordinates are aligned. As shown in Fig.1, for a more effective representation of the UAV's dynamics model, $F_I = \{O_i X_i Y_i Z_i\}$ represents the ground coordinate system, $F_B = \{O_b X_b Y_b Z_b\}$ represents the UAV body frame coordinate system, and $F_C = \{O_c X_c Y_c Z_c\}$ represents the camera frame coordinate system.

In order to characterize rotational motion, this article introduce the rotation matrix $R \in SO(3)$, which belongs to the set $\{R \in \mathbb{R}^{3\times 3} \mid \det R = 1, RR^T = R^T R = I_3\}$, where $SO(3)$ denotes a collection of specialized orthogonal matrices. The rotation matrix that establishes the correlation between the UAV body frame coordinate and the ground coordinate, utilizing Euler angles, is ascertainable by adhering to the methodology delineated by:

$$R(\gamma) = \begin{bmatrix} c\theta c\psi & s\theta c\psi s\phi - s\psi c\phi & s\theta c\psi c\phi + s\psi s\phi \\ c\theta s\psi & s\theta s\psi s\phi + c\psi c\phi & s\theta s\psi c\phi - c\psi s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \quad (1)$$

where $\gamma = [\phi, \theta, \psi]^T$, ϕ, θ, ψ respectively represent the roll angle, pitch angle, and yaw angle of the UAV. $s(\cdot) \triangleq \sin(\cdot)$, $c(\cdot) \triangleq \cos(\cdot)$.

Owing to the the rotation matrix between UAV body frame coordinate system and the camera frame coordinate system, and for the sake of ease in computation, the coordinate transformation between the UAV and the camera merely requires straightforward translation calculation for the distance between the UAV and objects. Subsequently, the dynamics equations for the position and attitude of the UAV are expressed by the following formulas:

$$\dot{p} = v_b, \quad (2)$$

$$\dot{v_b} = -g\hat{e}_3 + \frac{T}{m} R\hat{e}_3 + d_e, \quad (3)$$

$$\Omega\dot{\gamma} = \omega \quad (4)$$

$$J\dot{\omega} = -\omega^{\times} J\omega + \tau + d_\tau \quad (5)$$

where $p \in \mathbb{R}^3$ represents the UAV's center position of gravity expressed in F_I, $v_b \in \mathbb{R}^3$ denotes linear velocity, g represents for gravitational acceleration, m represents mass, $T \in \mathbb{R}$ represents thrust in the UAV frame, and $\hat{e}_3 \triangleq [0, 0, 1]^T$ represents the unit vector along the z-axis in coordinate system I. $d_e \in \mathbb{R}^3$ includes unmodeled dynamics and external disturbances and represents perturbation accelerations in the X_b, Y_b, and Z_b directions. $\omega = [\omega_\phi, \omega_\theta, \omega_\psi]^T$ represents the angular velocity and the $\times$ represents $\xi^{\times} = [0, -\xi_3, \xi_2;\ \xi_3, 0, -\xi_1;\ -\xi_2, \xi_1, 0]$. $J \in \mathbb{R}^{3\times 3}$ represents the internal matrix, $\tau \in \mathbb{R}^3$ represents the applied torque, $d_\tau \in \mathbb{R}^3$ represents the disturbance torque and $\Omega = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & s\phi c\theta \\ 0 & -s\phi & c\phi c\theta \end{bmatrix}$.

B. Control Objective

The control objective is to devise a controller that can be employed in the condition where velocity and position information of the target is not available, but can effectively guide the image feature towards the desired value. The motion of target can be obtained by depth camera. Consider the dynamics equations (2)-(5). Therefore, the approach can be accomplished stably with information related to image features, relative position, and velocity.

The tracking platform primarily consists of two critical components: the visual object tracking unit and the image-based visual servo control unit. The considered IBVS control is divided into two subtasks: visual servo model (Section III-A) and the design of tracking controller (Section III-C).

III. Main Result

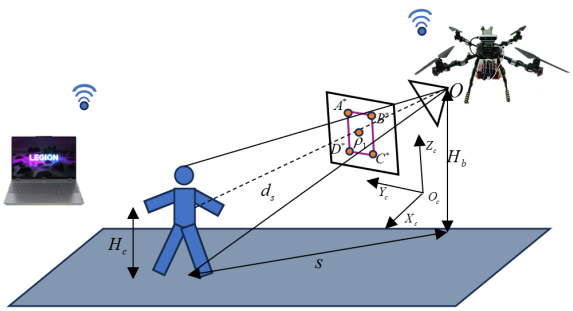
In this section, a methodology is proposed for estimating the relative position between the object and the UAV. Subsequently, considering the dynamics characteristics of the UAV

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on March 30,2026 at 23:06:09 UTC from IEEE Xplore. Restrictions apply.

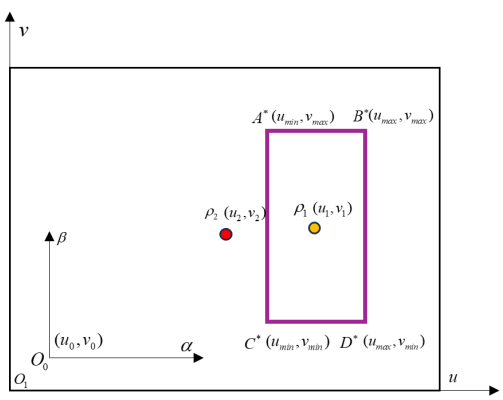
as well as internal and external disturbances, a robust tracking controller is devised to overcome the challenge of acquiring real-time measurements of the current distance and velocity, which are subsequently utilized as inputs to the tracker.

A. IBVS Model

The visual servo component, illustrated in Fig.3, represents the IBVS control approach and consists of two sections. Leveraging the outcomes of the YOLOv3 algorithm, we introduce an IBVS-based control approach to accomplish target tracking.



(a) Tracking platform



(b) Pixel coordinate and physical coordinate

Fig. 2: Relative estimation between an object and UAV

Define the image frame as illustrated in Fig.2(b). In this frame, (u, v) and (α, β) respectively represent the image frame in pixel and physical units with origins at O_1 and O_0. Define dx and dy as the physical dimensions of each pixel along the X and Y axes, with units of $\mathrm{m/pixel}$. The coordinates of any pixel in the image frame are interrelated as follows:

$$\begin{cases} u = \frac{\alpha}{dx} + u_0 \\ v = \frac{\beta}{dy} + v_0 \end{cases} \tag{6}$$

where (u_0, v_0) represents the origin physical coordinate of the image frame. (6) can be transformed into homogeneous coordinate matrix form, which is shown as follows:

$$\begin{bmatrix} \alpha \\ \beta \\ 1 \end{bmatrix} = \begin{bmatrix} dx & 0 & -u_0 dx \\ 0 & dy & -v_0 dy \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{7}$$

According to principles of imaging, the pinhole imaging model is commonly employed to approximate the position of any point in space onto the image plane. The projected position of any spatial point onto the image plane is often referred to as the line denoted as $o\rho_1$, as illustrated in Fig.2(a). The point of intersection between the optical center o and ρ_1 on the image plane is referred to as a perspective projection. Drawing inspiration from [8], the following relation is expressed as:

$$\begin{cases} \alpha_1 &= \frac{l y_1}{x_1} \\ \beta_1 &= \frac{l z_1}{x_1} \end{cases} \tag{8}$$

where l represents a constant, (α_1, β_1) represents the image physical coordinate of ρ_1, and (x_1, y_1, z_1) denotes the coordinate of spatial point ρ_1 in the camera coordinate system F_C. Perspective projection (8) can be transformed into the following matrix:

$$s_x \begin{bmatrix} \alpha_1 \\ \beta_1 \\ 1 \end{bmatrix} = \begin{bmatrix} l & 0 & 0 \\ 0 & l & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ z_1 \\ x_1 \end{bmatrix} \tag{9}$$

where the variable s_x represents a scale factor of (9). According to formulas (7) and (9), it can be deduced that:

$$\begin{aligned} s_x \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} &= \begin{bmatrix} \frac{1}{dx} & 0 & u_0 \\ 0 & \frac{1}{dy} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} l & 0 & 0 \\ 0 & l & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ z_1 \\ x_1 \end{bmatrix} \\ &= \begin{bmatrix} \alpha_x & 0 & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ z_1 \\ x_1 \end{bmatrix} \end{aligned} \tag{10}$$

where $\alpha_x = l/d_x$ and $\alpha_y = l/d_y$ represent the normalized focal lengths on the horizontal axis and vertical axis, respectively. (u_1, v_1) represents the coordinates of point ρ_1 in the image frame. Furthermore, the parameters $(\alpha_x, \alpha_y, u_0, v_0)$ are solely associated with the camera's internal characteristics.

Relative Position Estimation: In the execution of the IBVS task, it is essential to calculate the target's relative position and velocity from the UAV within the inertial coordinate system. Therefore, we present a feasible estimation approach and initially outline the subsequent assumptions. Suppose that the height of object is relatively stable, and the height of the UAV's camera is equal to the position of the target center point, as the mission progresses.

Remark III.1. *The assumption that object height remains relatively stable is well-founded, because it applies to common tracking entities like rigid bodies, vehicles, and stationary pedestrians[10]. Equating the height of camera and target center point is feasible, because the UAV reaches the target center quickly. This approach enables the UAV to neglect the impact of pitch during flight, simplify subsequent calculations.*

Define depth distance as d_s. When utilizing a depth camera, it not only captures pixel data but also provides precise depth distance information. Under appropriate lighting conditions, object depth distance is acquired through depth information.

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on March 30,2026 at 23:06:09 UTC from IEEE Xplore. Restrictions apply.

The algorithm for calculating depth information through deep learning methods has been integrated into the depth image from depth camera nodes. Therefore, we only need to invoke it and devise the algorithm to calculate the depth distance.

The object's distance from the camera's desired distance, represented as s^*, is determined by practical considerations. s represents the distance between camera and target in X_c axis in F_C. For the purposes of attitude control, where our primary manipulation is the yaw angle, we can safely neglect minor variations in roll as well as the slight variation in altitude between the camera and the object's center during flight. Therefore, consider the calculated depth distance to be approximately equal to the actual distance to the target, i.e., $s \approx d_s$. The algorithm 1 presents the acquisition of d_s:

Algorithm 1 obtain the depth distance d_s

Require: the target captured by camera
Ensure: the control of UAV
  detect the target and segment the image into $random$
  calculate $\rho(u, v)$
  define the depth search range min_{val};
  while $random$ **do**
    generate random noise bias within the range:
    $bias = (-min_{val}, +min_{val})$
    obtain depth information:
    $dist = data[(mid_{pos}[1] + bias), (mid_{pos}[0] + bias)]$
    array holds each depth information:
    $distance.append(dist)$
    bubble sorting and median filtering on $distance$;
    $d_s =$ the average value of the $distance$
    return d_s
  end while

B. Problem Transformation

utilizing the perspective projection of the visual target onto the image plane[11], [12], the image features are derived to characterize the UAV's motion according to (10): $q_x = s$, $q_y = q_x u/a_x$, $q_z = q_x u/a_y$, where q_x, q_y, q_z represent the positions of target calculated by depth camera in F_C. Thus define: $q = [q_x, q_y, q_z]^T$, $\dot{q} = [\dot{q_x}, \dot{q_x}\dot{u}/a_x, \dot{q_x}\dot{v}/a_y]^T$. Utilizing 10 as the foundational framework and supposing the desired distance s^* in F_B to be known, we deduce the equations governing the deviations between the center point ρ_1 and the point ρ_2 in F_C and Fig.2. The derived equations are as follows:

$$\begin{aligned} \triangle q_x &= (s - s^*) \\ \triangle q_y &= \frac{s(u_1 - u_2)}{a_x} \\ \triangle q_z &= \frac{s(v_1 - v_2)}{a_y} \end{aligned} \tag{11}$$

Referring to sources (2)(3)(11), the equation governing the dynamics of relative position and velocity in F_B is given by:

$$\dot{\tilde{q}} = -\tilde{v}_q \tag{12}$$

$$\dot{\tilde{v}}_q = -g\hat{e}_3 + f + d_e + \frac{T}{m}(R(\gamma) - R(\gamma_c))\hat{e}_3 \tag{13}$$

where $\tilde{q} = [\triangle q_x, \triangle q_y, \triangle q_z]^T + q_t$, $q_t = [x^*, 0, z^*]^T$ represents the distance z, where x^* represents the vertical distance between the camera and the top bracket of the UAV, and z^* represents the horizontal distance between the camera and the front bracket of the UAV. $f = TR(\gamma_c)\hat{e}_3/m \triangleq [f_x, f_y, f_z]^T$, f represents the force controller. γ_c is the command Euler angle. For perturbation terms, $\lim_{\gamma_c \to \gamma} T(R(\gamma) - R(\gamma_c))\hat{e}_3/m \to 0$, the nominal dynamics model is represented as:

$$\dot{\tilde{q}} = -\tilde{v}_q \tag{14}$$

$$\dot{\tilde{v}}_q = -g\hat{e}_3 + f + d_e \tag{15}$$

Controlling linear velocity alone is insufficient. Thus, the control of ψ_b needs to be integrated to further elaborate upon a comprehensive modeling approach.

C. Design of Robust Error Tracker

1) force controller : In this section a reliable approach based on robust control is referenced by [4]. The main objective of developing the controller is to minimize the deviations ($\triangle q_x$, $\triangle q_y$, $\triangle q_z$) and drive them towards zero. To tackle the challenges posed by unpredictable disturbances in real-world scenarios, a dynamics estimator has been integrated:

$$\dot{E}_v = -\epsilon_t(E_v - \tilde{v}_q) \tag{16}$$

$$\dot{E}_u = -\epsilon_t(E_u - g\hat{e}_3 + f) \tag{17}$$

where the initial values are set as $E_v(0) = 0$, $E_u(0) = 0$, $\epsilon_t > 0$ represents a constant parameter. The state estimator can be devised as follows:

$$\hat{u}_a = -\epsilon_t(E_v - \tilde{v}_q) + E_u \tag{18}$$

Filtering both sides of (15), and combining with (16) and (17), result in:

$$\dot{E}_v = -E_u + \hat{u}_a \tag{19}$$

Define a new variable:

$$\eta = \tilde{v}_q + \Gamma_1 \tilde{q} \tag{20}$$

where $\Gamma_1 = \mathrm{diag}\{\gamma_{1x}, \gamma_{1y}, \gamma_{1z}\}$ represents a positive definite matrix with positive constants γ_{1x} , γ_{1y}, γ_{1z}. The vector f in (15) is mentioned as the force controller:

$$f = g\hat{e}_3 - \Gamma_2(\tanh(\eta) + \tanh(\tilde{v}_q)) - sat_{\bar{u}}(\hat{u}_a) \tag{21}$$

where $\Gamma_2 = \mathrm{diag}\{\gamma_{2x}, \gamma_{2y}, \gamma_{2z}\}$ represents a diagonal matrix with positive constants and $\bar{u}$ represents the boundary condition. Substituting (21) into (15), the closed-loop position dynamics is expressed as follows:

$$\dot{\tilde{v}}_q = -\Gamma_2(\tanh(\eta) + \tanh(\tilde{v}_q)) - \tilde{u}_a^* \tag{22}$$

where $\tilde{u}_a^* = sat_{\bar{u}}(\hat{u}_a) - d_e$

Theorem III.1. *The force controller* (21) *is devised based on the nominal error dynamics model* (14) *and* (15)*. Considering* $\Gamma_{2z} < (g - \bar{u})/2$ *as the boundary for control parameters, the error system of closed-loop position is uniformly ultimately bounded.*

Proof. Refer to [4] for the detailed proof. □

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on March 30,2026 at 23:06:09 UTC from IEEE Xplore. Restrictions apply.

2) torque controller: The attitude control is defined as:

$$\triangle\psi_b = \frac{u_1 - u_2}{\delta} \tag{23}$$

where $\triangle\psi_b$ represents the yaw angle in UAV body frame and u_2 represents the abscissa of image coordinate center ρ_2 in Fig.2(b). The constant δ is a positive scalar. According to (23), a PID controller is devised and utilized to drive the convergence of $\triangle\psi_b$ to zero, as follows:

$$\omega_\psi(t) = K_P\triangle\psi_b(t) + K_I\int_0^t \triangle\psi_b(t)d_t + K_D\frac{d\triangle\psi_b(t)}{dt} \tag{24}$$

where K_p, K_I, K_D represent PID control parameters. d_t represents the time variable, and $\triangle\psi_b(t)$ denotes the error between the spatial point ρ_1 and point ρ_2 in Fig.2(b). Thus the result is obtained as the angular velocity of yaw.

The UAV dynamics model is described by (2) and (3). If the target tracking conditions fulfill the assumptions, then the target tracking can be achieved by the force controller (22) and the torque controller (24).

D. Method Flowchart

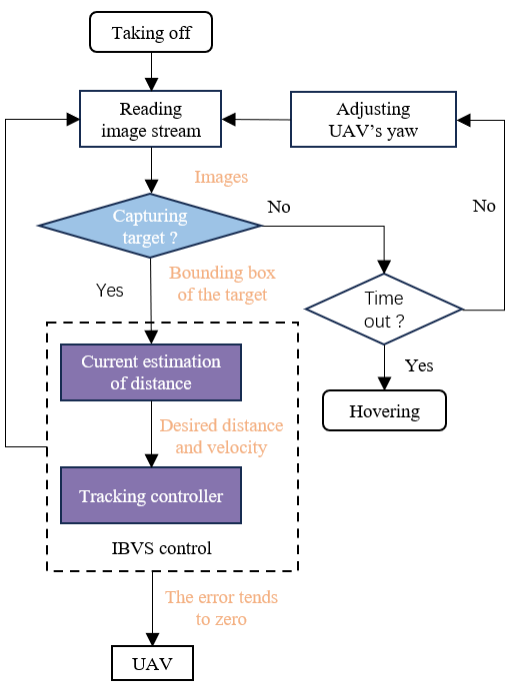


Fig. 3: Flowchart of the proposed object tracking platform.

Fig. 3 provides an illustration of the entire procedure to enhance understanding. Initially, camera information is transmitted to the ground station via a network protocol to assist the UAV in recognizing the target's position. Subsequently, the image recognition problem is transformed into a target tracking challenge. Then, a robust control strategy based on dynamics estimators is applied and simulated. Finally, a series of flight experiments, which includes simulations and practical tests, is conducted using an actual UAV to validate the proposed approach.

IV. EXPERIMENT

In this section, we construct a Gazebo simulation platform based on XTdrone [13] and conduct initial validation of the controller. Furthermore, we perform practical experiments to verify the feasibility of our controller.

A. Simulation

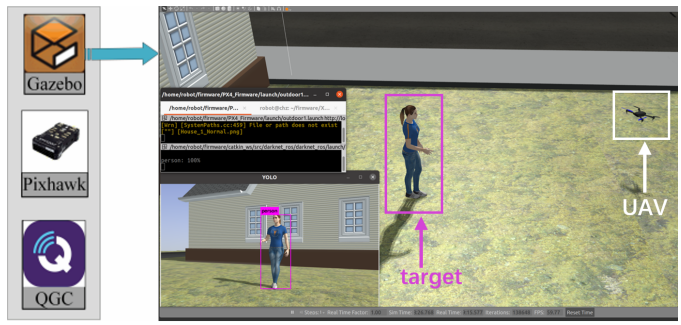


Fig. 4: The simulation scenario and platform

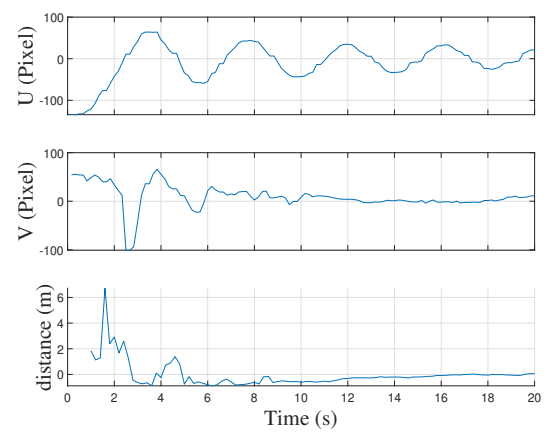


Fig. 5: The target tracking simulation comprises measuring the distance from the center point of the image and calculating the error from the desired distance and depth distance.

1) Simulation condition: The control simulation platform consists of three essential components: Gazebo, PX4-Autopilot, and QGroundControl (QGC), as depicted in Fig.4. Gazebo serves as a simulator renowned for its capacity to simulate robot populations within complex environments. The PX4-autopilot platform integrates the simulation environment within Gazebo, while QGC is responsible for facilitating communication between the simulation platform and the ground.

2) Simulation and results: The steps to accomplish the simulation are as follows: First, initiate the PX4 Autopilot simulation to launch the virtual UAV in Gazebo. Next, launch QGC to verify if the UAV is connected to the host computer. Subsequently, launch the virtual UAV and position the pedestrian within the UAV's FOV. Finally, execute the IBVS based on robust control file. To evaluate the performance of our simulation platform, we record the aerial data of the UAV in Gazebo. In this experiment, the desired distance between the target and the UAV is 4m. The parameters of simulation Γ_1 and Γ_2 are given as $\mathrm{diag}\{2, 1, 0.05\}$,$\mathrm{diag}\{0.4, 0.2, 0.005\}$

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on March 30,2026 at 23:06:09 UTC from IEEE Xplore. Restrictions apply.

respectively. Fig.5 shows the tracking trajectory under this condition, which shows the UAV fluently detects and identifies where the pedestrian is and gradually achieve the tracking task in the sixth second. Although there is a subtle transformation in simulation, it remains within an acceptable range owing to the feedback control theory.

B. Practical Experiment

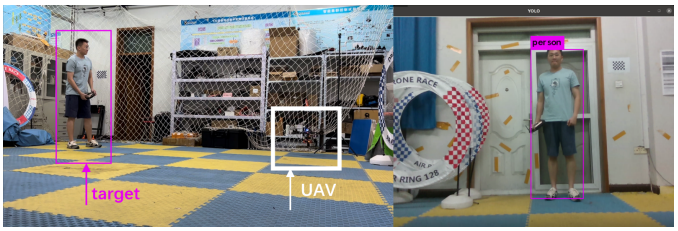


Fig. 6: The physical scenario and platform
(Check the hyperlink of experiment for more details)

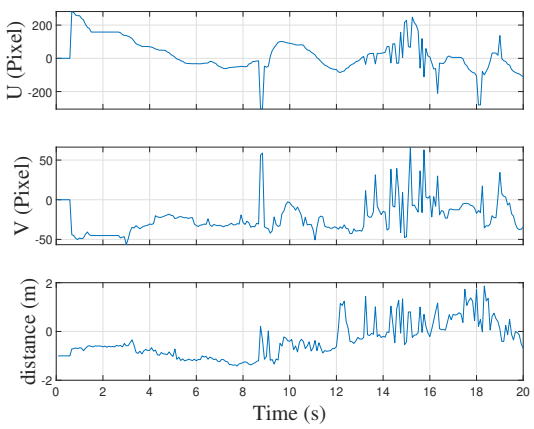


Fig. 7: The target tracking platform includes distance from the center point of the image and errors in both the desired distance and depth distance measurement

To validate its real-world utility, we conduct practical experiments. The process of launching the physical platform is similar to that in simulation but the transformation of image information is a little different. Owing to gazebo's Integration, the image information and transformation . But in actual situation by using the Intel NUC as UAV's computer, which cannot process image information well. Thus, we utilize the depth camera node to transmit the image stream to the ground computer, which runs the target detection algorithm and sends the target's coordinates in the image frame to the UAV's computer.

As depicted in Fig.6, the UAV and the target pedestrian are positioned at a desired distance of 3.5 meters. The camera captures images at a resolution of 640×480 pixels. The experiment's parameters, denoted as Γ_1 and Γ_2, are set as $\mathrm{diag}\{2, 2, 0.05\}$ and $\mathrm{diag}\{0.6, 4, 0.01\}$ respectively. Additionally, $a_x = a_y = 616.765$, $u_0 = 318.482$, $v_0 = 241.167$, $\epsilon_t = 0.5$, and $\bar{u} = 0.5$. Fig.7 illustrates the distance over time between the pedestrian's center point ρ_1 and the image coordinate center point ρ_2, with time represented on the x-axis. The pedestrian can be seen to be tracked smoothly and gradually, which is almost achieved at the ninth second. However, we observe fluctuations in the UAV's flight data around the expected values after the ninth second. This is because the pedestrian began to continuously circulate within the experimental area. Although the movement of the target causes the UAV to experience some shaking, it is still able to accurately identify the pedestrian's position. This effectively validates the robustness of the experiment.

Drawing from the aforementioned observations and visual data, it becomes increasingly apparent that the UAV demonstrates a notable degree of self-stabilization capability and adeptly tracks the target over time through the utilization of visual servo based on robust control.

V. Conclusion

This paper presents an IBVS approach to address the tracking problem for UAVs in challenging scenarios where GPS is unavailable, and it is not reliable to obtain the position and velocity of the target. Specifically, it calculates relative position and velocity using robust control based on image information and depth distance, and integrates attitude control for real-time tracking. This approach is validated as effective through simulations and experiments.

References

[1] Vanegas, F., Campbell, D., Roy, N., Gaston, K. J., & Gonzalez, F. (2017). UAV tracking and following a ground target under motion and localisation uncertainty. In 2017 IEEE Aerospace Conference (pp. 1-10). IEEE.

[2] Qin, C., Yu, Q., Go, H. H., & Liu, H. H. T. (2023). Perception-Aware image-based visual servoing of aggressive quadrotor UAVs. IEEE/ASME Transactions on Mechatronics.

[3] Yang, L., Liu, Z., Wang, G., & Wang, X. (2020). Image-Based visual servo control for ground target tracking using a fixed-wing UAV with pan-tilt camera. In 2020 International Conference on Unmanned Aircraft Systems (ICUAS) (pp. 354-361). IEEE.

[4] Xia, K., Shin, M., Chung, W., Kim, M., Lee, S., & Son, H. (2022). Landing a quadrotor UAV on a moving platform with sway motion using robust control. Control Engineering Practice, 128, 105288.

[5] Chaumette, F., & Hutchinson, S. (2006). Visual servo control. i. basic approaches. IEEE Robotics & Automation Magazine, 13(4), 82-90.

[6] Guenard, N., Hamel, T., & Mahony, R. (2008). A practical visual servo control for an unmanned aerial vehicle. IEEE Transactions on Robotics, 24(2), 331-340.

[7] Shi, H., Li, X., Hwang, K. S., Pan, W., & Xu, G. (2016). Decoupled visual servoing with fuzzy Q-learning. IEEE Transactions on Industrial Informatics, 14(1), 241-252.

[8] Miranda-Moya, A., Castañeda, H., Gordillo, J. L., & Wang, H. (2022). IBVS based on adaptive sliding mode control for a quadrotor target tracking under perturbations. Mechatronics, 88, 102909.

[9] J. Redmon, "Darknet: open source neural networks in c. 2013–2016".

[10] Liu, Y., Meng, Z., Zou, Y., & Cao, M. (2021). Visual object tracking and servoing control of a nano-scale quadrotor: system, algorithms, and experiments. IEEE CAA J. Autom. Sinica, 8(2), 344-360.

[11] Serra, P., Cunha, R., Hamel, T., Cabecinhas, D., & Silvestre, C. (2016). Landing of a quadrotor on a moving target using dynamic image-based visual servo control. IEEE Transactions on Robotics, 32(6), 1524-1535.

[12] Zheng, D., Wang, H., Chen, W., & Wang, Y. (2017). Planning and tracking in image space for image-based visual servoing of a quadrotor. IEEE Transactions on Industrial Electronics, 65(4), 3376-3385.

[13] Xiao, K., Tan, S., Wang, G., An, X., Wang, X., & Wang, X. (2020, June). XTDrone: A customizable multi-rotor UAVs simulation platform. In 2020 4th International Conference on Robotics and Automation Sciences (ICRAS) (pp. 55-61). IEEE.

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on March 30,2026 at 23:06:09 UTC from IEEE Xplore. Restrictions apply.